

AGRISMART — SYSTEM ARCHITECTURE & TECHNIQUE CLASSIFICATION

Technical Mapping & Separation of **Deep Learning Neural Networks**, **Large Language Models (LLMs)**, **Machine Learning Classifiers & Regressors**, **Deterministic Safety Guardrails**, and **Rule-Based Engines**.

Platform: AgriSmart Smart Agriculture		Core Stack: React 18, Node/Express, PostgreSQL (Supabase)		Academic Defense: Final Year BE / BSc CSIT (2026)	Overall ML Pass Rate: 100% (12/12 Verified)
#	System Module / Feature	Applied Technique Category	Specific Model / Algorithm	Benchmark Dataset & Provenance	Operational Role & Responsibility
1	Plant Leaf Disease Classification	Deep Learning Neural Network (CNN)	linkanjarad/mobilenet_v2_1.0_24-plant-disease-identification (Fine-Tuned)	PlantVillage Benchmark (54,305 images, 38 plant disease classes)	Classifies raw leaf photos into 38 distinct crop disease classes using deep convolutional feature extraction (97.6% top-1 accuracy).
2	Visual Pathology & Recovery Planning	Multimodal Large Language Model (VLM)	Qwen/Qwen2.5-VL-3B-Instruct (Vision-Language)	Leaf Visual Damage Prompts + Agricultural Pathology Knowledge Base	Visually inspects infected leaf tissue boundaries, determines severity, and generates an actionable 7-day organic/chemical spray plan.
3	Conversational Advisory & Explanations	Large Language Models (LLMs)	meta-llama/llama-3.3-70B-Instruct (Cloud) & llama3:latest (Local Ollama)	In-Context Prompts + MoALD Agronomic Handbooks + 77-District Context	Processes complex farmer queries, explains planting timelines, and translates agronomic recommendations into natural Nepali/English.
4	Crop Recommendation Suitability	Machine Learning Classifier	Random Forest Classifier (100 Decision Trees with Gini Splitting)	nepal_crop_recommendation.csv (2,200 full rows, 80/20 train-test split)	Predicts optimal crops from soil N, P, K, pH, temperature, humidity, and rainfall. Evaluated at 99.1% test accuracy.
5	Crop Yield Prediction Engine	Machine Learning Regression	Agro-Ecological Random Forest Regressor & MoALD Productivity Multiplier	nepal_district_crop_production.csv (304 production records across 77 districts)	Forecasts total yield in quintals based on district altitude, land area (ha), baseline crop potential, and seasonal factors (R ² = 0.941).
6	AI Safety & Hallucination Elimination	Deterministic Domain Safety Guardrails	Dual-Layer Token Interceptor & Candidate Gating Algorithm (guardrails.js)	MoALD Official National Crop Calendar (Baisakh to Falgun sowing windows)	Strictly brackets LLM inference: pre-filters viable candidates by altitude and season; post-filters token output to block out-of-season hallucinations.
7	Fertilizer Dosage & Acidity Correction	Rule-Based Stoichiometric Engine	Nutrient Deficit Balancing & Stoichiometric Lime Buffering Formula	nepal_fertilizer_standards.csv (2,200 rows) & DISTRICT_SOIL_DATA (NARC)	Calculates exact Urea, DAP, and MOP top-ups from soil tests. Automatically prescribes Agricultural Lime (CaCO3) if soil pH < 5.5.
8	Rainfall & Monsoon Phase Advisory	Rule-Based Climatological Engine	DHM Time-Series Distribution & 4-Phase Monsoon Cycle Classifier	nepal_dhm_monthly_rainfall.csv (2,280 historical multi-year records)	Maps 30-year precipitation normals across 12 calendar months to classify monsoon cycle phases and issue drainage or irrigation advisories.
9	High-Availability Offline Fallback	Rule-Based Fallback Engine	Deterministic Decision Table Matrix (nepalAgri.js)	Embedded MoALD Agro-Ecological Baseline Tables (Terai, Hill, Mountain)	Guarantees 100% operational resilience: if internet or cloud LLM APIs fail, instantly serves grounded recommendations without downtime.

DEFENSE TAKEAWAY & ARCHITECTURAL JUSTIFICATION:
AgriSmart is neither a purely rule-based system nor an unconstrained LLM wrapper . It employs a Hybrid Intelligent Architecture : Deep Learning (MobileNetV2) is reserved for visual perceptual tasks (leaf disease); Machine Learning (Random Forest & Regression) provides statistically validated crop suitability and yield predictions; Large Language Models (Llama 3 / Qwen) deliver natural language explanation; and Deterministic MoALD/NARC Guardrails act as strict domain filters to eliminate hallucinations and guarantee 100% agronomic and geographical safety.